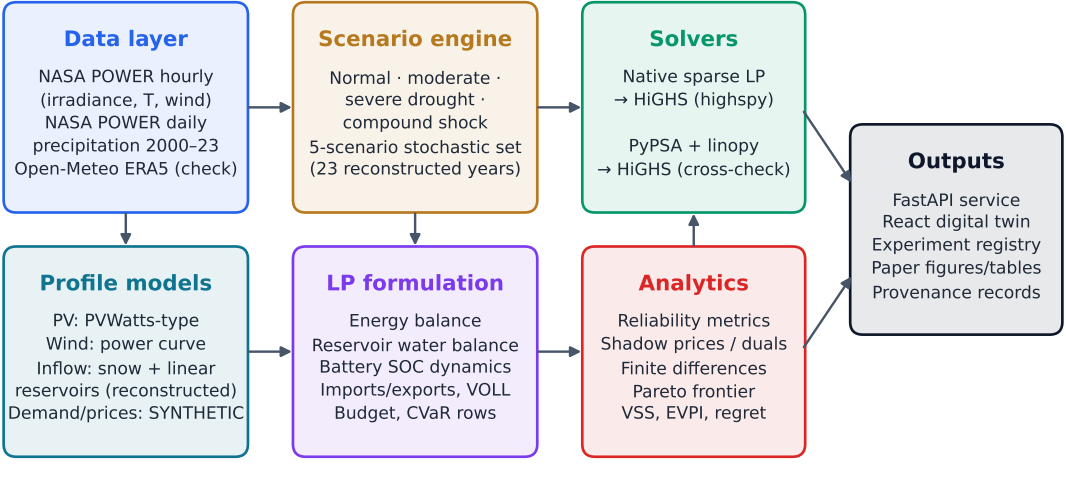
Data layer
NASA POWER hourly (irradiance, T, wind)
NASA POWER daily precipitation 2000–23
Open-Meteo ERA5 (check)
Scenario engine
Normal · moderate · severe drought · compound shock
5-scenario stochastic set (23 reconstructed years)
Solvers
Native sparse LP → HiGHS (highspy)
PyPSA + linopy → HiGHS (cross-check)
Outputs
FastAPI service
React digital twin
Experiment registry
Paper figures/tables
Provenance records
Profile models
PV: PVWatts-type
Wind: power curve
Inflow: snow + linear reservoirs (reconstructed)
Demand/prices: SYNTHETIC
LP formulation
Energy balance
Reservoir water balance
Battery SOC dynamics
Imports/exports, VOLL
Budget, CVaR rows
Analytics
Reliability metrics
Shadow prices / duals
Finite differences
Pareto frontier
VSS, EVPI, regret